

School of Computer Science and Engineering

B.Tech (CSE) Semester

Project ID- BTP2CSE315

Project Title- Anomaly-Based Network Traffic Monitoring System

Team Members-

Anshu rana

Anunay pandey

Imran Ansari

Nitin kumar

Rohit Singh

• Outlook of Presentation

- Abstract
- Introduction
- Literature
- Implementation
- Results
- Conclusion
- Research Paper
- Publication

Abstract

- ▶ The rapid growth of network-based services has increased traffic volume and complexity, making networks more vulnerable to security threats.
- ▶ Anomalous activities such as DoS and DDoS attacks can severely impact the availability and performance of network services.
- ▶ This project focuses on identifying deviations from normal traffic patterns using anomaly-based detection techniques.
- ▶ By establishing a baseline of normal network behavior, the system continuously monitors traffic data to detect irregular activities.
- ▶ Detection methods include Z-score thresholding and K-Means clustering, achieving detection accuracy above 91%.
- ▶ The system enables early detection of anomalies, reducing response time and supporting network administrators in maintaining secure operations.

Introduction

- **Growing Complexity**

Cloud workloads, IoT deployments, and remote-first work models have pushed more traffic onto networks, increasing both volume and variety of potential threats.

- **Rising Threats**

DoS and DDoS attacks have become routine. Recent DDoS campaigns have crossed terabit-per-second thresholds, overwhelming well-provisioned defenses.

- **Limitations of Traditional IDS**

Signature-based systems cannot detect novel attacks, zero-day exploits, or low-and-slow intrusions — they only recognize previously known patterns.

- **Anomaly-Based Approach**

Instead of matching known-bad patterns, this system asks: "does this traffic look normal?" — learning behavioral baselines and flagging deviations in real time.

Literature Review

1987

Denning — Foundations of Anomaly Detection

Proposed statistical profiles of normal user behavior and flagging of significant deviations — the conceptual foundation of this project.

1999

Roesch — Snort — Signature-Based IDS

Popularized signature-based detection. High precision on known attacks but fundamentally limited by the need to keep signatures current.

2004

Lakhina et al. — PCA on Network Flows

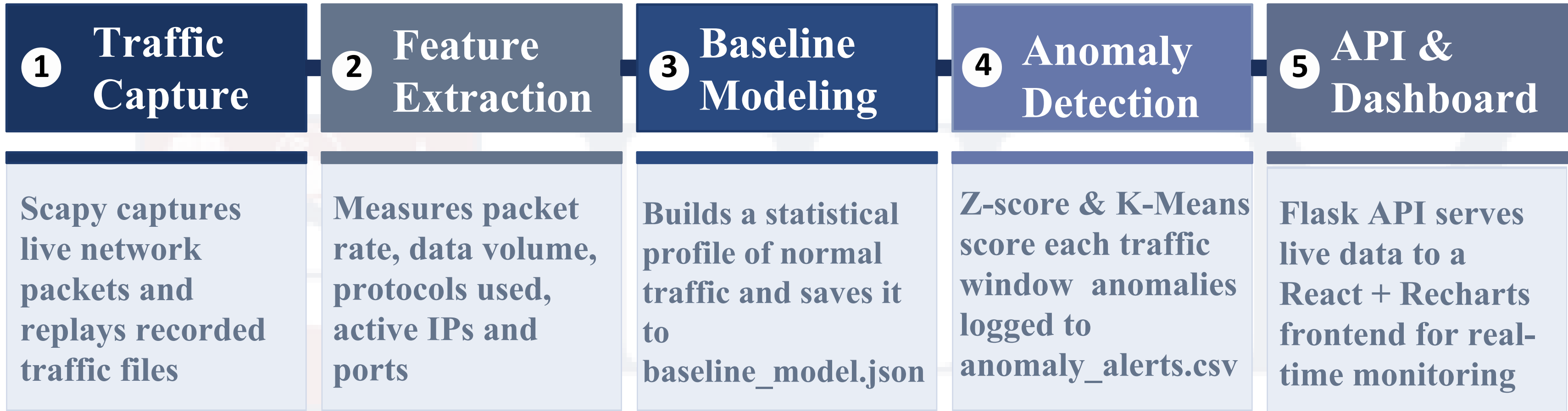
Showed Principal Component Analysis on flow-level traffic features exposes network-wide anomalies missed by per-link monitoring.

2016

Bontemps et al. — LSTM for DDoS Detection

Demonstrated LSTM networks capture temporal flow structure well enough for high-accuracy DDoS detection.

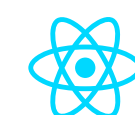
Implementation



High-Level Purpose

- The system monitors network traffic and detects anomalous behavior such as:
 - DoS, DDoS, Port Scan, Brute Force, SYN Flood, DNS Amplification
- It combines:
 - statistical baseline detection, K-Means clustering
- It also provides:
 - alert generation, logging, a React-based dashboard

Tools

-  React, Recharts, Vite, ESLint
-  Framer Motion
-  Scapy, Pandas,
-  NumPy, Scikit-learn, Matplotlib

Results

- Live data captured from NetGuard — Anomaly Monitor

5,088,910

Packets / Sec

+12.4% live traffic

403

Active Connections

+3.2% from baseline

237

Anomalies (24h)

Peak hour at 14:00

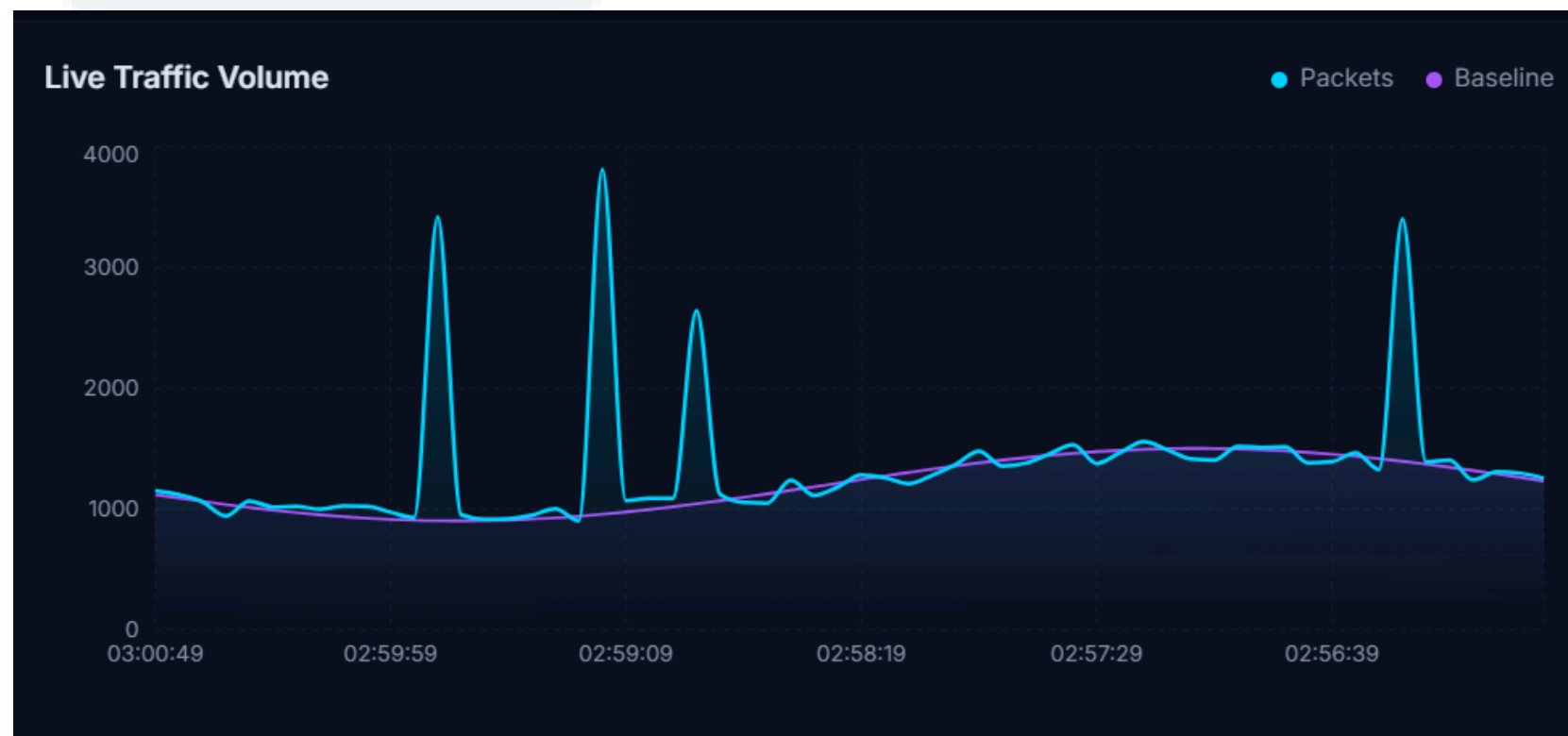
91%+

Detection Accuracy

Z-score + K-Means engine

Dashboard Features:

1. Real-time traffic monitoring
2. Alert management
3. Detection results visualization
4. System metrics and settings



Conclusion

1. **Effective Detection -**

The system successfully detected DoS and DDoS attacks with over 91% accuracy using a dual-method approach combining Z-score thresholding and K-Means clustering.

2. **Adaptive Baseline -**

EWMA-based dynamic baseline modeling allows the system to track legitimate behavioral shifts without triggering false positives.

3. **Real-Time Performance -**

Sub-100ms per-window processing latency confirms the system is lightweight enough for real-time deployment without dedicated hardware.

4. **Contribution**

Demonstrates a practical, low-cost anomaly detection system appropriate for small-to-medium networks, bridging the gap between research and deployment.

Paper Title -

- Anomaly-Based Network Traffic Monitoring System

Author -

- Anshu rana , Imran Ansari , Anunay Pandey , Rohit Kumar , Nitin Singh
- SCSE, IILM University

Abstract Contribution

- Networks are increasingly vulnerable to threats like DoS & DDoS attacks that disrupt performance.
- The system monitors traffic, establishes a normal baseline, & detects deviations automatically.
- Using Z-score and K-Means clustering, it achieves over 91% detection accuracy.

Key Sections Covered

- Abstract, Introduction, Literature Review, Implementation
- Traffic Capture, Feature Extraction, Baseline Modeling, Anomaly Detection, API & Dashboard
- Results & Evaluation, Conclusion, Research Paper, Publication

Publication

Conference Name:

Track:

- Computer Networks & Cybersecurity
- **Paper Title:** Anomaly-Based Network Traffic Monitoring System
- **Paper ID:**
- **Authors:** Anshu rana , Anunay Pandey , Imran Ansari , Rohit kumar , Nitin Singh
- **Submission Date:**
- **Status:**

Thank You